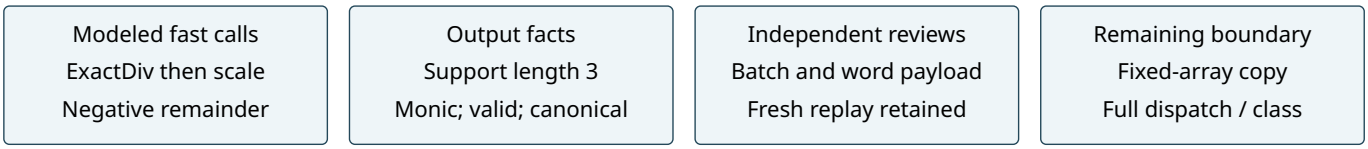
Fast polynomial-call endpoint and independent word reviews

The new producer endpoint constructs the modeled fast branch through exact division, literal word scaling and the final negative remainder. Two independent predecessor reviews separately confirm the batch and coefficient/payload components. The new endpoint still awaits its own review.



Call construction. Input lists Ua=[ra,qa,1], Ub=[rb,qb,1] and trimmed Va=[ta,sa] decode to the stated polynomials. Actual slope words feed the existing multiplication/addition list models in native order. Proved residual divisibility constructs a successful exactDiv return. Cancellation by the nonzero input product identifies its decoded quotient with the field quotient, rather than assuming that equality. Actual PegcdWord.scale with the computed inverseLead then yields a canonical, normalized monic support of degree two. The modeled remainder of pneg(V) supplies canonical normalized ordinate output and decoded Mumford validity. Output support has length three and last word one; the remainder has at most two words, supplying the final list-size preconditions.

Producer checks. Five main declarations, five fixed controls and ten named readbacks passed with unchanged bindings; the false theorem was rejected. The complete F17 control invokes the exactDiv/scale/remainder endpoint with actual residual witnesses and constructs existential return lists. It is not a separate numeric Rust run. Two expression-normalization failures are retained without weakened hypotheses. Prime 2<p<2^31, canonical coefficient and f words, actual nonzero norm/slope guards, degree-five f and both residual congruences remain explicit.

Independent batch review. At commit 5937d99b29, the non-author reviewer freshly rebuilt Wide and Inversion alongside the three batch modules, then checked ten producer readbacks, seven controls, three added boundary controls and four bridge readbacks. The false theorem and three malformed checker inventories were rejected. Native order and ni/si orientation match the restricted source dispatch without assuming an inverse witness. A premature checker invocation returned 127 while rustc was running; the same compiler completed, after which the checker passed. This harness failure is retained separately from theorem verification.

Independent coefficient/payload review. At commit 6d7317d5ea, eight producer modules and 27 readbacks passed fresh replay. Three reviewer fixture declarations bring the total to 30 readbacks; nine checker mutations and the false theorem were rejected. Input indices, reversed ordinate differences, operation order, guard timing, S0 boundary and inverse-leading sign match. The arithmetic sample p17 gives N2,S1=16,S0=7, ni9,si16, slope(12,8), inverseLead13. The separate full residual fixture has slope(0,16), inverseLead16 and quotient -X^2+X before scaling. A fixture launch with the wrong Lean module-root directory failed before elaboration and is retained; correcting the directory passed without source changes.

Domain finding. The p=3 controls prove the arithmetic statements at that boundary. Native Field.validate rejects odd p below five for study admission. Arithmetic theorem coverage must therefore not be presented as benchmark admission. This distinction does not require weakening the field or word arithmetic theorems.

Custody and independence. Both reviewers report machine passed and faithful scoped correspondence for their named predecessor claims. They authored neither assigned component and report no sibling-report access; historical context was inherited, not blind re-derived. Deeper helper and library caches were reused and content-bound. The batch review preserves one generic unmatched helper outside the actual import closure, with its inventory and exact hashes. Executable caches are omitted from this archive; source, raw logs, mappings, failures and large artifact-store binding files are retained. No automated independence-checker pass is claimed.

Next composition. The endpoint establishes modeled list outputs, not the physical fixed-array copy or compiled Rust behavior. Next review the exact call chain, prove the array copy/decode contract and compose full dispatch; fast/fallback output equality and divisor-class semantics remain distinct. No scientific execution, timing or cost result occurred. The frozen 1,260-target scope and campaign limits are unchanged.